

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2004

Blountsville 138 kV
December 02, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2004. The Interconnection Customer proposes a 54.4 MW battery energy storage facility interconnecting at the Blountsville 138 kV substation in the Pennsylvania area. The request was received on February 09, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 518 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 20 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,440,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Marquis – Friendly Hills 138 kV L
Contingency: Ritter Park – Sugar Hill 345 kV Line
Loading: 154.4% (Rating: 254 MVA)
DFAX: 9.0%
Pre-project loading: 152.4%
2. Facility: Sugar Hill Substation – Guntersville Dam 138 kV
Contingency: Ritter Park – Sugar Hill 345 kV
Loading: 154.1% (Rating: 217 MVA)
DFAX: 9.0%
Pre-project loading: 151.8%
3. Facility: Belfield Substation – Guntersville Dam 138 kV L
Contingency: Ritter Park – Sugar Hill 345 kV
Loading: 150.1% (Rating: 241 MVA)
DFAX: 9.0%
Pre-project loading: 148.1%
4. Facility: Friendly Hills – Uc Dais 138 kV
Contingency: Sugar Hill – Ritter Park 345 kV
Loading: 146.9% (Rating: 278 MVA)
DFAX: 9.0%
Pre-project loading: 145.2%
5. Facility: Belfield – Marquis 138 kV
Contingency: Sugar Hill – Ritter Park 345 kV
Loading: 144.2% (Rating: 261 MVA)
DFAX: 9.0%
Pre-project loading: 142.3%

6. Facility: Maljamar – Coopers Mills 138 kV line
Contingency: Lighthouse Hill – Aes Beaver Valley Partners Beaver Valley 345 kV
Loading: 139.5% (Rating: 208 MVA)
DFAX: 8.5%
Pre-project loading: 137.3%
7. Facility: Uc Davis – Ritter Park 138 kV
Contingency: Ritter Park – Sugar Hill 345 kV L
Loading: 136.3% (Rating: 312 MVA)
DFAX: 9.0%
Pre-project loading: 134.8%
8. Facility: Coopers Mills – Eagle 138 kV
Contingency: Aes Beaver Valley Partners Beaver Valley – Lighthouse Hill 345 kV
Loading: 131.1% (Rating: 229 MVA)
DFAX: 8.5%
Pre-project loading: 129.1%
9. Facility: Stewart Mtn Dam – Shepaug 138 kV
Contingency: Sugar Hill – Ritter Park 345 kV L
Loading: 127.4% (Rating: 390 MVA)
DFAX: 17.1%
Pre-project loading: 125.1%
10. Facility: Kirtland Air Force Base 4 – Maljamar 138 kV
Contingency: Lighthouse Hill – Aes Beaver Valley Partners Beaver Valley 345 kV
Loading: 127.2% (Rating: 219 MVA)
DFAX: 8.5%
Pre-project loading: 125.1%
11. Facility: Valley – Potash Jct 138 kV
Contingency: Aes Beaver Vly Partners Beaver Vly – Lighthouse Hill 345 kV
Loading: 122.3% (Rating: 207 MVA)
DFAX: 8.5%
Pre-project loading: 120.1%
12. Facility: Vly – Eagle 138 kV
Contingency: Lighthouse Hill – Aes Beaver Valley Partners Beaver Vly 345 kV
Loading: 122.3% (Rating: 201 MVA)
DFAX: 8.5%
Pre-project loading: 120.0%
13. Facility: Yucca – Fraley 138 kV
Contingency: Sugar Hill – Allegheny Energy Units 3 4 & 5 345 kV
Loading: 119.2% (Rating: 383 MVA)
DFAX: 5.8%
Pre-project loading: 118.4%
14. Facility: Fraley – Houseville Substation 138 kV Line
Contingency: Sugar Hill – Allegheny Energy Units 3 4 & 5 345 kV
Loading: 119.1% (Rating: 391 MVA)
DFAX: 5.8%
Pre-project loading: 118.3%
15. Facility: Va Hospital – Houseville 138 kV
Contingency: Sugar Hill – Allegheny Energy Units 3 4 & 5 345 kV
Loading: 119.0% (Rating: 395 MVA)
DFAX: 5.8%
Pre-project loading: 118.2%
16. Facility: Sugar Hill – Va Hospital 138 kV L
Contingency: Sugar Hill – Allegheny Energy Units 3 4 & 5 345 kV

Loading: 118.9% (Rating: 403 MVA)

DFAX: 5.8%

Pre-project loading: 118.1%

17. Facility: N – Shepaug 138 kV line

Contingency: Ritter Park – Sugar Hill 345 kV

Loading: 113.4% (Rating: 429 MVA)

DFAX: 17.1%

Pre-project loading: 111.3%

18. Facility: Kirtland Air Force Base 4 – Lighthouse Hill Substation 138 kV Line

Contingency: Aes Beaver Vly Partners Beaver Valley – Lighthouse Hill 345 kV

Loading: 109.3% (Rating: 237 MVA)

DFAX: 8.5%

Pre-project loading: 107.3%

19. Facility: Raccoon – Septa Substation 138 kV

Contingency: Aes Beaver Valley Partners Beaver Valley – Lighthouse Hill 345 kV

Loading: 100.8% (Rating: 239 MVA)

DFAX: 6.0%

Pre-project loading: 99.4%

20. Facility: Blountsville Substation – Vista 138 kV

Contingency: Stewart Mtn Dam 345/138 kV Transformer

Loading: 100.2% (Rating: 208 MVA)

DFAX: 27.6%

Pre-project loading: 93.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Marquis – Friendly Hills 138 kV L	Rebuild line with higher-capacity conductor	\$2,597,000	38% (\$980,000)
Sugar Hill Substation – Guntersville Dam 138 kV	Rebuild line with higher-capacity conductor	\$2,861,000	38% (\$1,080,000)
Belfield – Guntersville Dam 138 kV	Rebuild line with higher-capacity conductor	\$2,229,000	38% (\$841,000)
Friendly Hills – Uc Dais 138 kV	Rebuild line with higher-capacity conductor	\$2,335,000	38% (\$881,000)
Belfield – Marquis 138 kV	Rebuild line with higher-capacity conductor	\$3,102,000	38% (\$1,171,000)
Maljamar – Coopers Mills 138 kV line	Rebuild line with higher-capacity conductor	\$3,210,000	5% (\$155,000)
Uc Davis Substation – Ritter Park 138 kV	Rebuild line with higher-capacity conductor	\$1,834,000	38% (\$692,000)
Coopers Mills – Eagle 138 kV	Rebuild line with higher-capacity conductor	\$4,974,000	5% (\$239,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Stewart Mtn Dam – Shepaug 138 kV	Rebuild line with higher-capacity conductor	\$2,475,000	17% (\$432,000)
Kirtland Air Force Base 4 – Maljamar 138 kV	Rebuild line with higher-capacity conductor	\$3,376,000	5% (\$163,000)
Vly – Potash Jct Substation 138 kV	Rebuild line with higher-capacity conductor	\$2,627,000	4% (\$109,000)
Vly – Eagle 138 kV	Rebuild line with higher-capacity conductor	\$1,383,000	4% (\$58,000)
Yucca – Fraley 138 kV	Rebuild line with higher-capacity conductor	\$2,931,000	10% (\$285,000)
Faley – Houseville 138 kV	Rebuild line with higher-capacity conductor	\$4,799,000	10% (\$467,000)
Houseville – Va Hospital 138 kV L	Rebuild line with higher-capacity conductor	\$2,428,000	10% (\$236,000)
Sugar Hill – Va Hospital 138 kV	Rebuild line with higher-capacity conductor	\$4,912,000	10% (\$478,000)
N Substation – Shepaug 138 kV	Rebuild line with higher-capacity conductor	\$4,363,000	17% (\$761,000)
Lighthouse Hil – Kirtland Air Force Base 4 138 kV line	Reconductor line	\$878,000	5% (\$42,000)
Raccoon – Septa 138 kV	Reconductor line	\$2,605,000	23% (\$590,000)
Vista – Blountsville 138 kV	Reconductor line	\$1,541,000	51% (\$780,000)
	Total allocated to project		\$10,440,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.